Module 24

Relational Database Design/4

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March 17, 2026

Module Recap

- Introduced the theory of functional dependencies

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- Discussed issues in 'good' design in the context of functional dependencies

Module Objectives

Goals

- To Learn Algorithms for Properties of Functional Dependencies

Module Outline

- Algorithms for Functional Dependencies

Algorithms for Functional Dependencies

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- **Computing closure of F**
  - For each $\gamma \subseteq R$, we find the closure γ^+, and for each $S \subseteq \gamma^+$, we output a functional dependency $\gamma \rightarrow S$.

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- Example: Given $F = \{A \rightarrow C, AB \rightarrow CD\}$
  - C is extraneous in $AB \rightarrow CD$ since $AB \rightarrow C$ can be inferred even after deleting $C \circ (AB)^+ = ABCD$ in $\{A \rightarrow C, AB \rightarrow D\}$

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- For example: on RHS: $\{A \rightarrow B, B \rightarrow C, A \rightarrow CD\}$ can be simplified to $\{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$

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- For example: on RHS: $\{A \rightarrow B, B \rightarrow C, A \rightarrow CD\}$ can be simplified to $\{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$
  - In the forward: (1) $A \rightarrow CD \Rightarrow A \rightarrow C$ and $A \rightarrow D$ (2) $A \rightarrow B, B \rightarrow C \Rightarrow A \rightarrow C$

Canonical Cover (2): Example

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- For example: $A \rightarrow C$ is redundant in: $\{A \rightarrow B, B \rightarrow C, A \rightarrow C\}$
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  - In the reverse: (1) $A \rightarrow B, B \rightarrow C \Rightarrow A \rightarrow C$ (2) $A \rightarrow C, A \rightarrow D \Rightarrow A \rightarrow CD$
- For example: on LHS: $\{A \rightarrow B, B \rightarrow C, AC \rightarrow D\}$ can be simplified to $\{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$

Canonical Cover (2): Example

- For example: $A \rightarrow C$ is redundant in: $\{A \rightarrow B, B \rightarrow C, A \rightarrow C\}$
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- For example: $A \rightarrow C$ is redundant in: $\{A \rightarrow B, B \rightarrow C, A \rightarrow C\}$
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  - In the reverse: $A \rightarrow D \Rightarrow AC \rightarrow D$

Canonical Cover (3): RHS

- $\{A \rightarrow B, B \rightarrow C, A \rightarrow CD\} \Rightarrow \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$

Canonical Cover (3): RHS

- $\{A \rightarrow B, B \rightarrow C, A \rightarrow CD\} \Rightarrow \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$
  - (1) $A \rightarrow CD \Rightarrow A \rightarrow C$ and $A \rightarrow D$

Canonical Cover (3): RHS

- $\{A \to B, B \to C, A \to CD\} \Rightarrow \{A \to B, B \to C, A \to D\}$
  - (1) $A \to CD \Rightarrow A \to C$ and $A \to D$
  - (2) $A \to B, B \to C \Rightarrow A \to C$

Canonical Cover (3): RHS

- $\{A \to B, B \to C, A \to CD\} \Rightarrow \{A \to B, B \to C, A \to D\}$
  - (1) $A \to CD \Rightarrow A \to C$ and $A \to D$
  - (2) $A \to B, B \to C \Rightarrow A \to C$
  - $A^+ = ABCD$

Canonical Cover (3): RHS

- $\{A \to B, B \to C, A \to CD\} \Rightarrow \{A \to B, B \to C, A \to D\}$
  - (1) $A \to CD \Rightarrow A \to C$ and $A \to D$
  - (2) $A \to B, B \to C \Rightarrow A \to C$
  - $A^+ = ABCD$
- $\{A \to B, B \to C, A \to D\} \Rightarrow \{A \to B, B \to C, A \to CD\}$

Canonical Cover (3): RHS

- $\{A \to B, B \to C, A \to CD\} \Rightarrow \{A \to B, B \to C, A \to D\}$
  - (1) $A \to CD \Rightarrow A \to C$ and $A \to D$
  - (2) $A \to B, B \to C \Rightarrow A \to C$
  - $A^+ = ABCD$
- $\{A \to B, B \to C, A \to D\} \Rightarrow \{A \to B, B \to C, A \to CD\}$
  - $A \to B, B \to C \Rightarrow A \to C$

Canonical Cover (3): RHS

- $\{A \to B, B \to C, A \to CD\} \Rightarrow \{A \to B, B \to C, A \to D\}$
  - (1) $A \to CD \Rightarrow A \to C$ and $A \to D$
  - (2) $A \to B, B \to C \Rightarrow A \to C$
  - $A^+ = ABCD$
- $\{A \to B, B \to C, A \to D\} \Rightarrow \{A \to B, B \to C, A \to CD\}$
  - $A \to B, B \to C \Rightarrow A \to C$
  - $A \to C, A \to D \Rightarrow A \to CD$

Canonical Cover (3): RHS

- $\{A \to B, B \to C, A \to CD\} \Rightarrow \{A \to B, B \to C, A \to D\}$
  - (1) $A \to CD \Rightarrow A \to C$ and $A \to D$
  - (2) $A \to B, B \to C \Rightarrow A \to C$
  - $A^+ = ABCD$
- $\{A \to B, B \to C, A \to D\} \Rightarrow \{A \to B, B \to C, A \to CD\}$
  - $A \to B, B \to C \Rightarrow A \to C$
  - $A \to C, A \to D \Rightarrow A \to CD$
  - $A^+ = ABCD$

Canonical Cover (4): LHS

- $\{A \rightarrow B, B \rightarrow C, AC \rightarrow D\} \Rightarrow \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$

Canonical Cover (4): LHS

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Canonical Cover (4): LHS

- $\{A \rightarrow B, B \rightarrow C, AC \rightarrow D\} \Rightarrow \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$
  - $A \rightarrow B, B \rightarrow C \Rightarrow A \rightarrow C \Rightarrow A \rightarrow AC$
  - $A \rightarrow AC, AC \rightarrow D \Rightarrow A \rightarrow D$

Canonical Cover (4): LHS

- $\{A \to B, B \to C, AC \to D\} \Rightarrow \{A \to B, B \to C, A \to D\}$
  - $A \to B, B \to C \Rightarrow A \to C \Rightarrow A \to AC$
  - $A \to AC, AC \to D \Rightarrow A \to D$
  - $A^+ = ABCD$

Canonical Cover (4): LHS

- $\{A \to B, B \to C, AC \to D\} \Rightarrow \{A \to B, B \to C, A \to D\}$
  - $A \to B, B \to C \Rightarrow A \to C \Rightarrow A \to AC$
  - $A \to AC, AC \to D \Rightarrow A \to D$
  - $A^+ = ABCD$
- $\{A \to B, B \to C, A \to D\} \Rightarrow \{A \to B, B \to C, AC \to D\}$

Canonical Cover (4): LHS

- $\{A \to B, B \to C, AC \to D\} \Rightarrow \{A \to B, B \to C, A \to D\}$
  - $A \to B, B \to C \Rightarrow A \to C \Rightarrow A \to AC$
  - $A \to AC, AC \to D \Rightarrow A \to D$
  - $A^+ = ABCD$
- $\{A \to B, B \to C, A \to D\} \Rightarrow \{A \to B, B \to C, AC \to D\}$
  - $A \to D \Rightarrow AC \to D$

Canonical Cover (4): LHS

- $\{A \to B, B \to C, AC \to D\} \Rightarrow \{A \to B, B \to C, A \to D\}$
  - $A \to B, B \to C \Rightarrow A \to C \Rightarrow A \to AC$
  - $A \to AC, AC \to D \Rightarrow A \to D$
  - $A^+ = ABCD$
- $\{A \to B, B \to C, A \to D\} \Rightarrow \{A \to B, B \to C, AC \to D\}$
  - $A \to D \Rightarrow AC \to D$
  - $AC^+ = ABCD$

Canonical Cover (5)

- To compute a canonical cover for F:

```
repeat
    Use the union rule to replace any dependencies in F
        alpha_1 -> beta_1 and alpha_1 -> beta_2
        with alpha_1 -> beta_1 beta_2
    Find a functional dependency alpha -> beta with an
    extraneous attribute either in alpha or in beta
        /* Note: test for extraneous attributes done using F_c, not F */
        If an extraneous attribute is found, delete it from alpha -> beta
until F does not change
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        /* Note: test for extraneous attributes done using F_c, not F */
        If an extraneous attribute is found, delete it from alpha -> beta
until F does not change
```

- Note: Union rule may become applicable after some extraneous attributes have been deleted, so it has to be re-applied

Canonical Cover (6): Example

- $R = (A, B, C)$

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- Combine $A \rightarrow BC$ and $A \rightarrow B$ into $A \rightarrow BC$

Canonical Cover (6): Example

- $R = (A, B, C)$
- $F = \{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$
- Combine $A \rightarrow BC$ and $A \rightarrow B$ into $A \rightarrow BC$
- Set is now $\{A \rightarrow BC, B \rightarrow C, AB \rightarrow C\}$

Canonical Cover (6): Example

- $R = (A, B, C)$
- $F = \{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$
- Combine $A \rightarrow BC$ and $A \rightarrow B$ into $A \rightarrow BC$
- Set is now $\{A \rightarrow BC, B \rightarrow C, AB \rightarrow C\}$
- A is extraneous in $AB \rightarrow C$

Canonical Cover (6): Example

- $R = (A, B, C)$
- $F = \{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$
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  - Check if the result of deleting A from $AB \rightarrow C$ is implied by the other dependencies

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- Set is now $\{A \rightarrow BC, B \rightarrow C, AB \rightarrow C\}$
- A is extraneous in $AB \rightarrow C$
  - ▷ Check if the result of deleting A from $AB \rightarrow C$ is implied by the other dependencies
  - ▷ ▷ Yes: in fact, $B \rightarrow C$ is already present!

Canonical Cover (6): Example

- $R = (A, B, C)$
- $F = \{A \to BC, B \to C, A \to B, AB \to C\}$
- Combine $A \to BC$ and $A \to B$ into $A \to BC$
- Set is now $\{A \to BC, B \to C, AB \to C\}$
- A is extraneous in $AB \to C$
  - ▷ Check if the result of deleting A from $AB \to C$ is implied by the other dependencies
  - ▷ ▷ Yes: in fact, $B \to C$ is already present!
- Set is now $\{A \to BC, B \to C\}$

Canonical Cover (6): Example

- $R = (A, B, C)$
- $F = \{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$
- Combine $A \rightarrow BC$ and $A \rightarrow B$ into $A \rightarrow BC$
- Set is now $\{A \rightarrow BC, B \rightarrow C, AB \rightarrow C\}$
- A is extraneous in $AB \rightarrow C$
  - ▷ Check if the result of deleting A from $AB \rightarrow C$ is implied by the other dependencies
  - ▷ ▷ Yes: in fact, $B \rightarrow C$ is already present!
- Set is now $\{A \rightarrow BC, B \rightarrow C\}$
- C is extraneous in $A \rightarrow BC$

Canonical Cover (6): Example

- $R = (A, B, C)$
- $F = \{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$
- Combine $A \rightarrow BC$ and $A \rightarrow B$ into $A \rightarrow BC$
- Set is now $\{A \rightarrow BC, B \rightarrow C, AB \rightarrow C\}$
- A is extraneous in $AB \rightarrow C$
  - ▷ Check if the result of deleting A from $AB \rightarrow C$ is implied by the other dependencies
  - ▷ ▷ Yes: in fact, $B \rightarrow C$ is already present!
- Set is now $\{A \rightarrow BC, B \rightarrow C\}$
- C is extraneous in $A \rightarrow BC$
  - ▷ Check if $A \rightarrow C$ is logically implied by $A \rightarrow B$ and the other dependencies

Canonical Cover (6): Example

- $R = (A, B, C)$
- $F = \{A \to BC, B \to C, A \to B, AB \to C\}$
- Combine $A \to BC$ and $A \to B$ into $A \to BC$
- Set is now $\{A \to BC, B \to C, AB \to C\}$
- A is extraneous in $AB \to C$
  - ▷ Check if the result of deleting A from $AB \to C$ is implied by the other dependencies
  - ▷ ▷ Yes: in fact, $B \to C$ is already present!
- Set is now $\{A \to BC, B \to C\}$
- C is extraneous in $A \to BC$
  - ▷ Check if $A \to C$ is logically implied by $A \to B$ and the other dependencies
  - ▷ ▷ Yes: using transitivity on $A \to B$ and $B \to C$.

Canonical Cover (6): Example

- $R = (A, B, C)$
- $F = \{A \to BC, B \to C, A \to B, AB \to C\}$
- Combine $A \to BC$ and $A \to B$ into $A \to BC$
- Set is now $\{A \to BC, B \to C, AB \to C\}$
- A is extraneous in $AB \to C$
  - Check if the result of deleting A from $AB \to C$ is implied by the other dependencies
  - ▷ Yes: in fact, $B \to C$ is already present!
- Set is now $\{A \to BC, B \to C\}$
- C is extraneous in $A \to BC$
  - Check if $A \to C$ is logically implied by $A \to B$ and the other dependencies
  - ▷ Yes: using transitivity on $A \to B$ and $B \to C$.
  - Can use attribute closure of A in more complex cases

Canonical Cover (6): Example

- $R = (A, B, C)$
- $F = \{A \to BC, B \to C, A \to B, AB \to C\}$
- Combine $A \to BC$ and $A \to B$ into $A \to BC$
- Set is now $\{A \to BC, B \to C, AB \to C\}$
- A is extraneous in $AB \to C$
  - Check if the result of deleting A from $AB \to C$ is implied by the other dependencies
  - ▷ Yes: in fact, $B \to C$ is already present!
- Set is now $\{A \to BC, B \to C\}$
- C is extraneous in $A \to BC$
  - Check if $A \to C$ is logically implied by $A \to B$ and the other dependencies
  - ▷ Yes: using transitivity on $A \to B$ and $B \to C$.
  - Can use attribute closure of A in more complex cases
- The canonical cover is: $A \to B, B \to C$

Practice Problems on Functional Dependencies

- Find if a given functional dependency is implied from a set of Functional Dependencies:

Practice Problems on Functional Dependencies

- Find if a given functional dependency is implied from a set of Functional Dependencies:
- a) For: $A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC$

Practice Problems on Functional Dependencies

- Find if a given functional dependency is implied from a set of Functional Dependencies:
- a) For: $A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC$
  - ▷ i) Check: $BCD \rightarrow H$

Practice Problems on Functional Dependencies

- Find if a given functional dependency is implied from a set of Functional Dependencies:
- a) For: $A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC$
  - i) Check: $BCD \rightarrow H$
  - ii) Check: $AED \rightarrow C$

Practice Problems on Functional Dependencies

- Find if a given functional dependency is implied from a set of Functional Dependencies:
- a) For: $A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC$
  - i) Check: $BCD \rightarrow H$
  - ii) Check: $AED \rightarrow C$
- b) For: $AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A$

Practice Problems on Functional Dependencies

- Find if a given functional dependency is implied from a set of Functional Dependencies:
- a) For: $A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC$
  - i) Check: $BCD \rightarrow H$
  - ii) Check: $AED \rightarrow C$
- b) For: $AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A$
  - i) Check: $CF \rightarrow DF$

Practice Problems on Functional Dependencies

- Find if a given functional dependency is implied from a set of Functional Dependencies:
- a) For: $A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC$
  - i) Check: $BCD \rightarrow H$
  - ii) Check: $AED \rightarrow C$
- b) For: $AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A$
  - i) Check: $CF \rightarrow DF$
  - ii) Check: $BG \rightarrow E$

Practice Problems on Functional Dependencies

- Find if a given functional dependency is implied from a set of Functional Dependencies:
- a) For: $A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC$
  - i) Check: $BCD \rightarrow H$
  - ii) Check: $AED \rightarrow C$
- b) For: $AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A$
  - i) Check: $CF \rightarrow DF$
  - ii) Check: $BG \rightarrow E$
  - iii) Check: $AF \rightarrow G$

Practice Problems on Functional Dependencies

- Find if a given functional dependency is implied from a set of Functional Dependencies:
- a) For: $A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC$
  - i) Check: $BCD \rightarrow H$
  - ii) Check: $AED \rightarrow C$
- b) For: $AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A$
  - i) Check: $CF \rightarrow DF$
  - ii) Check: $BG \rightarrow E$
  - iii) Check: $AF \rightarrow G$
  - iv) Check: $AB \rightarrow EF$

Practice Problems on Functional Dependencies

- Find if a given functional dependency is implied from a set of Functional Dependencies:
- a) For: $A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC$
  - i) Check: $BCD \rightarrow H$
  - ii) Check: $AED \rightarrow C$
- b) For: $AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A$
  - i) Check: $CF \rightarrow DF$
  - ii) Check: $BG \rightarrow E$
  - iii) Check: $AF \rightarrow G$
  - iv) Check: $AB \rightarrow EF$
- c) For: $A \rightarrow BC, B \rightarrow E, CD \rightarrow EF$

Practice Problems on Functional Dependencies

- Find if a given functional dependency is implied from a set of Functional Dependencies:
- a) For: $A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD, DH \rightarrow BC$
  - i) Check: $BCD \rightarrow H$
  - ii) Check: $AED \rightarrow C$
- b) For: $AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A$
  - i) Check: $CF \rightarrow DF$
  - ii) Check: $BG \rightarrow E$
  - iii) Check: $AF \rightarrow G$
  - iv) Check: $AB \rightarrow EF$
- c) For: $A \rightarrow BC, B \rightarrow E, CD \rightarrow EF$
  - i) Check: $AD \rightarrow F$

Practice Problems on Functional Dependencies (2)

- Find Super Key using Functional Dependencies:

Practice Problems on Functional Dependencies (2)

- Find Super Key using Functional Dependencies:
- a) Relational Schema $R(ABCDE)$. Functional dependencies: $AB \rightarrow C, DE \rightarrow B, CD \rightarrow E$

Practice Problems on Functional Dependencies (2)

- Find Super Key using Functional Dependencies:
- a) Relational Schema $R(ABCDE)$. Functional dependencies: $AB \rightarrow C, DE \rightarrow B, CD \rightarrow E$
- b) Relational Schema $R(ABCDE)$. Functional dependencies: $AB \rightarrow C, C \rightarrow D, B \rightarrow EA$

Practice Problems on Functional Dependencies (3)

- Find Candidate Key using Functional Dependencies:

Practice Problems on Functional Dependencies (3)

- Find Candidate Key using Functional Dependencies:
- a) Relational Schema $R(ABCDE)$. Functional dependencies: $AB \rightarrow C, DE \rightarrow B, CD \rightarrow E$

Practice Problems on Functional Dependencies (3)

- Find Candidate Key using Functional Dependencies:
- a) Relational Schema $R(ABCDE)$. Functional dependencies: $AB \rightarrow C, DE \rightarrow B, CD \rightarrow E$
- b) Relational Schema $R(ABCDE)$. Functional dependencies: $AB \rightarrow C, C \rightarrow D, B \rightarrow EA$

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$
- b) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, C \rightarrow B\}$

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$
- b) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, C \rightarrow B\}$
- c) $R(ABCDEFGHIJ)$ having FDs $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$
- b) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, C \rightarrow B\}$
- c) $R(ABCDEFGHIJ)$ having FDs $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$
- d) $R(ABDLPT)$ having FDs $\{B \rightarrow PT, A \rightarrow D, T \rightarrow L\}$

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$
- b) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, C \rightarrow B\}$
- c) $R(ABCDEFGHIJ)$ having FDs $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$
- d) $R(ABDLPT)$ having FDs $\{B \rightarrow PT, A \rightarrow D, T \rightarrow L\}$
- e) $R(ABCDEFGH)$ having FDs $\{E \rightarrow G, AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A\}$

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$
- b) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, C \rightarrow B\}$
- c) $R(ABCDEFGHIJ)$ having FDs $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$
- d) $R(ABDLPT)$ having FDs $\{B \rightarrow PT, A \rightarrow D, T \rightarrow L\}$
- e) $R(ABCDEFGH)$ having FDs $\{E \rightarrow G, AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A\}$
- f) $R(ABCDE)$ having FDs $\{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$
- b) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, C \rightarrow B\}$
- c) $R(ABCDEFGHIJ)$ having FDs $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$
- d) $R(ABDLPT)$ having FDs $\{B \rightarrow PT, A \rightarrow D, T \rightarrow L\}$
- e) $R(ABCDEFGH)$ having FDs $\{E \rightarrow G, AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A\}$
- f) $R(ABCDE)$ having FDs $\{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$
- g) $R(ABCDEH)$ having FDs $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$
- b) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, C \rightarrow B\}$
- c) $R(ABCDEFGHIJ)$ having FDs $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$
- d) $R(ABDLPT)$ having FDs $\{B \rightarrow PT, A \rightarrow D, T \rightarrow L\}$
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- g) $R(ABCDEH)$ having FDs $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$
- **Prime Attributes**: Attribute set that belongs to any candidate key are called Prime Attributes

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$
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- c) $R(ABCDEFGHIJ)$ having FDs $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$
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- **Prime Attributes**: Attribute set that belongs to any candidate key are called Prime Attributes
- It is union of all the candidate key attribute: $\{CK_1 \cup CK_2 \cup CK_3 \cup \ldots\}$

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$
- b) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, C \rightarrow B\}$
- c) $R(ABCDEFGHIJ)$ having FDs $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$
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- g) $R(ABCDEH)$ having FDs $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$
- **Prime Attributes**: Attribute set that belongs to any candidate key are called Prime Attributes
- It is union of all the candidate key attribute: $\{CK_1 \cup CK_2 \cup CK_3 \cup \ldots\}$
- If Prime attribute determined by other attribute set, then more than one candidate key is possible.

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$
- b) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, C \rightarrow B\}$
- c) $R(ABCDEFGHIJ)$ having FDs $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$
- d) $R(ABDLPT)$ having FDs $\{B \rightarrow PT, A \rightarrow D, T \rightarrow L\}$
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- **Prime Attributes**: Attribute set that belongs to any candidate key are called Prime Attributes
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- If Prime attribute determined by other attribute set, then more than one candidate key is possible.
- For example, If A is Candidate Key, and $X \rightarrow A$, then, X is also Candidate Key.

Practice Problems on Functional Dependencies (4)

- Find Prime and Non Prime Attributes using Functional Dependencies:
- a) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F\}$
- b) $R(ABCDEF)$ having FDs $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, C \rightarrow B\}$
- c) $R(ABCDEFGHIJ)$ having FDs $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$
- d) $R(ABDLPT)$ having FDs $\{B \rightarrow PT, A \rightarrow D, T \rightarrow L\}$
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- f) $R(ABCDE)$ having FDs $\{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$
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- **Prime Attributes**: Attribute set that belongs to any candidate key are called Prime Attributes
- It is union of all the candidate key attribute: $\{CK_1 \cup CK_2 \cup CK_3 \cup \ldots\}$
- If Prime attribute determined by other attribute set, then more than one candidate key is possible.
- For example, If A is Candidate Key, and $X \rightarrow A$, then, X is also Candidate Key.
- **Non Prime Attribute**: Attribute set does not belong to any candidate key are called Non Prime Attributes

Practice Problems on Functional Dependencies (5)

- Check the Equivalence of a Pair of Sets of Functional Dependencies:

Practice Problems on Functional Dependencies (5)

- Check the Equivalence of a Pair of Sets of Functional Dependencies:
- a) Consider the two sets F and G with their FDs as below:

Practice Problems on Functional Dependencies (5)

- Check the Equivalence of a Pair of Sets of Functional Dependencies:
- a) Consider the two sets F and G with their FDs as below:
  - i) $F : A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H$

Practice Problems on Functional Dependencies (5)

- Check the Equivalence of a Pair of Sets of Functional Dependencies:
- a) Consider the two sets F and G with their FDs as below:
  - i) $F : A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H$
  - ii) $G : A \rightarrow CD, E \rightarrow AH$

Practice Problems on Functional Dependencies (5)

- Check the Equivalence of a Pair of Sets of Functional Dependencies:
- a) Consider the two sets F and G with their FDs as below:
  - i) $F : A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H$
  - ii) $G : A \rightarrow CD, E \rightarrow AH$
- b) Consider the two sets P and Q with their FDs as below:

Practice Problems on Functional Dependencies (5)

- Check the Equivalence of a Pair of Sets of Functional Dependencies:
- a) Consider the two sets F and G with their FDs as below:
  - i) $F : A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H$
  - ii) $G : A \rightarrow CD, E \rightarrow AH$
- b) Consider the two sets P and Q with their FDs as below:
  - i) $P : A \rightarrow B, AB \rightarrow C, D \rightarrow ACE$

Practice Problems on Functional Dependencies (5)

- Check the Equivalence of a Pair of Sets of Functional Dependencies:
- a) Consider the two sets F and G with their FDs as below:
  - i) $F : A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H$
  - ii) $G : A \rightarrow CD, E \rightarrow AH$
- b) Consider the two sets P and Q with their FDs as below:
  - i) $P : A \rightarrow B, AB \rightarrow C, D \rightarrow ACE$
  - ii) $Q : A \rightarrow BC, D \rightarrow AE$

Practice Problems on Functional Dependencies (6)

- Find the Minimal Cover or Irreducible Sets or Canonical Cover of a Set of Functional Dependencies:

Practice Problems on Functional Dependencies (6)

- Find the Minimal Cover or Irreducible Sets or Canonical Cover of a Set of Functional Dependencies:
- a) $AB \rightarrow CD, BC \rightarrow D$

Practice Problems on Functional Dependencies (6)

- Find the Minimal Cover or Irreducible Sets or Canonical Cover of a Set of Functional Dependencies:
- a) $AB \rightarrow CD, BC \rightarrow D$
- b) $ABCD \rightarrow E, E \rightarrow D, AC \rightarrow D, A \rightarrow B$

Module Summary

Summary

- Studied Algorithms for Properties of Functional Dependencies

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